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Augmented Reality for Reflective Robotic Welding Path Programming

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ABSTRACT

Robotic welding path programming is commonly framed as a technical problem addressed through expert-oriented tools. This pictorial presents a Research-through-Design (RtD) exploration of an Augmented Reality (AR) interface that allows operators to define seam and tack welding paths directly on physical workpieces using gestural input, while receiving immediate visual feedback on robotic reachability derived from inverse kinematics. This pictorial documents a series of design interactions in which system was progressively refined through deployment and designerly reflection, and examining how spatial interaction and infrastructural breakdowns can influence human understanding of robotic constraints during welding path planning. Through annotated images and reflective accounts of system use, this pictorial contributes (1) an AR-based path planning workflow that supports both seam and tack welding, and (2) design knowledge on how computational and visual feedback mediates embodied human-robot negotiation.

Authors Keywords

Augmented Reality (AR); Computational Design; Head-Mounted Display (HMD); Robotic Welding Path Planning

CSS Concepts

• Human-centered computing~Interaction design~Systems and tools for interaction design

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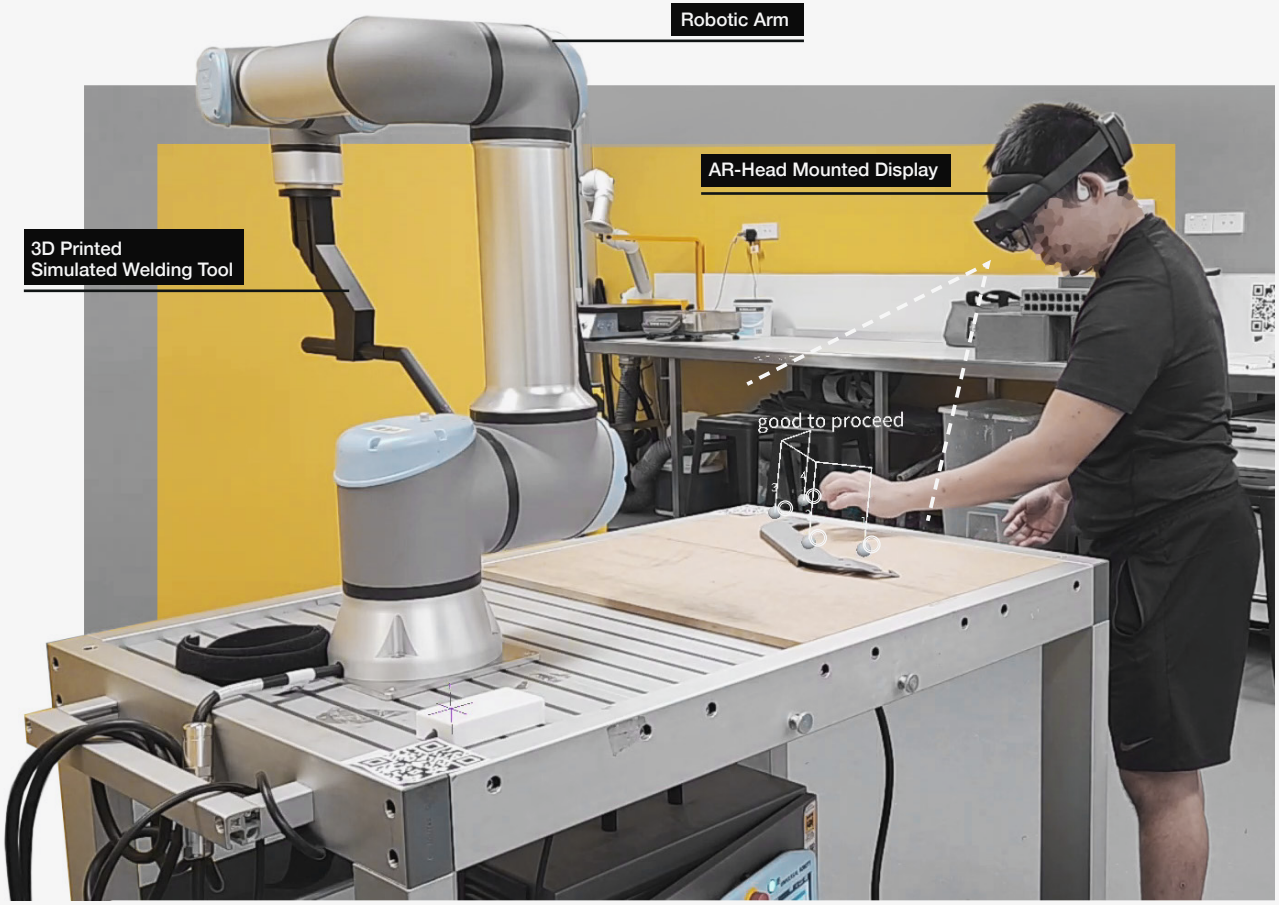


Figure 01. Mixed Reality snapshot of using the proposed AR-assisted HRI for robotic welding tack path planning. (source: Author, 2026)

PROBLEM SPACE

Welding is an essential, yet physically demanding, and hazardous profession that requires high levels of skill and spatial precision [8, 20]. Despite its critical role in manufacturing and construction settings, industries face a steady decline in the number of skilled welders, partly due to occupational health risks and the complexity of manual work [4]. In recent years, robotic welding systems have emerged as viable solutions to improve worker safety, productivity, and consistency while enabling mass customisation in fabrication [9,10].

However, the adoption of robotic welding remains limited. This challenge can be broadly divided into two core problems: path planning and execution. Given the inherent complexity involved in performing robotic welding tasks, this pictorial focuses exclusively on the former. Conventional programming methods, such as teach-pendant or CAD-based path definitions (refer to **Figure 2A**) often require expert knowledge and considerable effort to translate spatial intent into numerical parameters [22]. This translation of spatial intent into programming code is particularly challenging for designers and skilled welders who operate through embodied, situated understanding rather than parametric specifications.

In robotic welding, defining an accurate welding path is critical, as deviations in position or orientation can result in weak joints, incomplete fusion, or collisions between the robot and the workpiece. Because robotic operate within strict kinematic limits, even small changes in workpiece placement or path definition can cause the robotic arm to lose reachability or produce incorrect welds. These constraints make welding path planning not only a geometric task but also a spatial decision-making process that benefits from immediate visual or tactile feedback.

Augmented Reality (AR) offers an alternative mode of interaction that situates robotic programming directly within the physical workspace, as highlighted in [3,17,21,11]. By enabling users to visualise and manipulate robot configuration within the physical environment, these case studies showed AR capability as a spatially intuitive layer between human intent and robotic execution. Within the context of robotic welding, these AR systems are often associated with the Programming-through-Demonstration (PtD) paradigm (refer to **Figure 2B**), allowing users to trace continuous weld seams through tool motion [15,14,16].

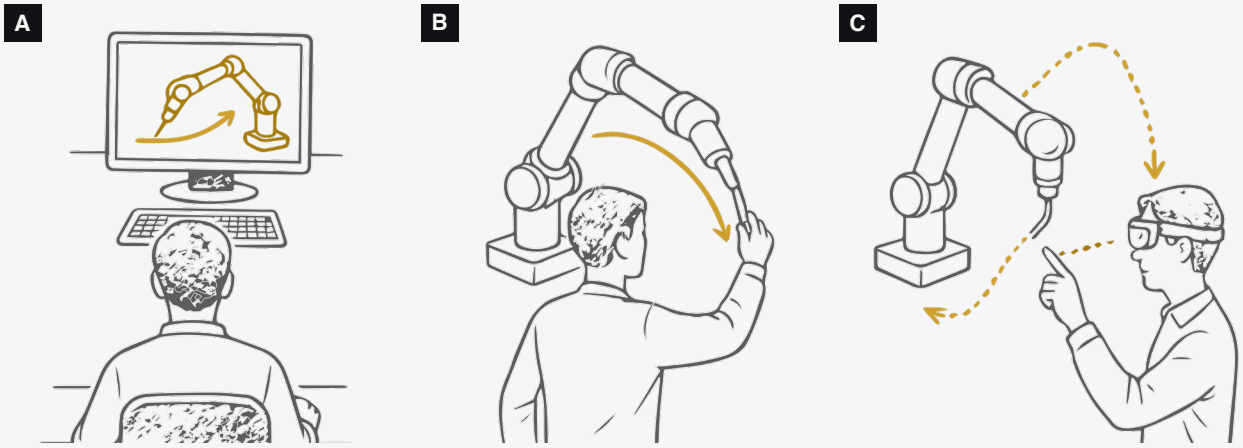


Figure 02. Different approaches to program robotic arm for welding task. (A) CAD-based approach; (B) Program-through-Demonstration (PtD); (C) AR-based programming. (source: Author, 2026)

However, these approaches provide limited support for tack welding, a crucial preparatory process that requires discrete spatial decisions and responsiveness to context. In welding practice, seam welding refers to a continuous joint along a surface, while tack welding consists of discrete weld points used to temporarily hold parts in place before final joining. Given that tack welding involves segmented placement rather than continuous motion, it requires frequent spatial judgement and adjustment, making it difficult to automate using conventional PtD approach. As a result, robotic support for tack welding remains relatively underexplored despite its significant role in manufacturing workflows.

This pictorial investigates how AR can extend robotic path programming to incorporate tack welding as a design-driven human-in-the-loop (HITL) activity. We iterative developed an AR interface that enables users to define both seam and tack welding paths through gestural-based spatial input while receiving real-time reachability feedback computed from the robot's inverse kinematics model. Rather than treating AR solely as a visualisation layer, we argue that the proposed system could position AR as an interactive medium which users can negotiate robotic constraints in situ. By situating computational feedback within a cyber-physical workspace, the system invites negotiation between user judgement, robotic constraint and material placement.

Our investigation adopts a Research-through-Design (RtD) approach to explore how AR feedback communicates robotic affordance and limitations (refer to **Figure 2C**). Each development iteration served as a probe for reflection, combining internal testing, video analysis and design journaling to explore how computational feedback mediate spatial decision-making. Rather than measuring usability performance, the pictorial aimed to generate design knowledge through reflective system development, specifically focusing on how AR modalities can support robotic path planning and facilitate reflective human-robot collaboration (HRC).

RESEARCH METHOD

This study adopts a Research-through-Design (RtD) methodology [23], positioning the development of an AR-assisted robotic welding interface as an exploratory design inquiry. RtD is suited to early-stage investigations where knowledge emerges through iterative making and reflection.

The investigation was conducted over six months, with each iteration developed and reflected upon sequentially. This designer-led study involved exclusively the author team (n = 3), who acted as both designers and evaluators; no external participants were involved. The research team consists solely of the three authors. The study comprised three design iterations, each conducted within a single structured session (three sessions in total). Each iteration functioned as a design probe, informing subsequent decisions as recurring interaction breakdowns emerged during path planning.

Reflection drew on multiple forms of design evidence, including screen recordings, annotated snapshots, reflective journals, and Post-it notes. Each iteration concluded with a group reflection session (three in total), where observations were documented and synthesised through thematic grouping, focusing on recurring breakdown events, interaction patterns, and design tensions across sessions. Reflection was further shaped by the differing demands of seam welding (continuous path definition) and tack welding (discrete spatial placement), informing both interaction design and interpretation.

Insights are articulated through three lenses: (1) system-level reflection, focusing on feedback clarity and AR stability; (2) interaction-level reflection, considering gesture-based path definition across welding tasks; and (3) contextual reflection, examining how AR visual cues mediate decision-making in robotic welding. These lenses were derived through iterative cross-comparison of observations rather than predefined categories. As a designer-led RtD study, the findings are not intended to be generalisable but to surface interactional qualities and design tensions for future AR-assisted HRI systems.

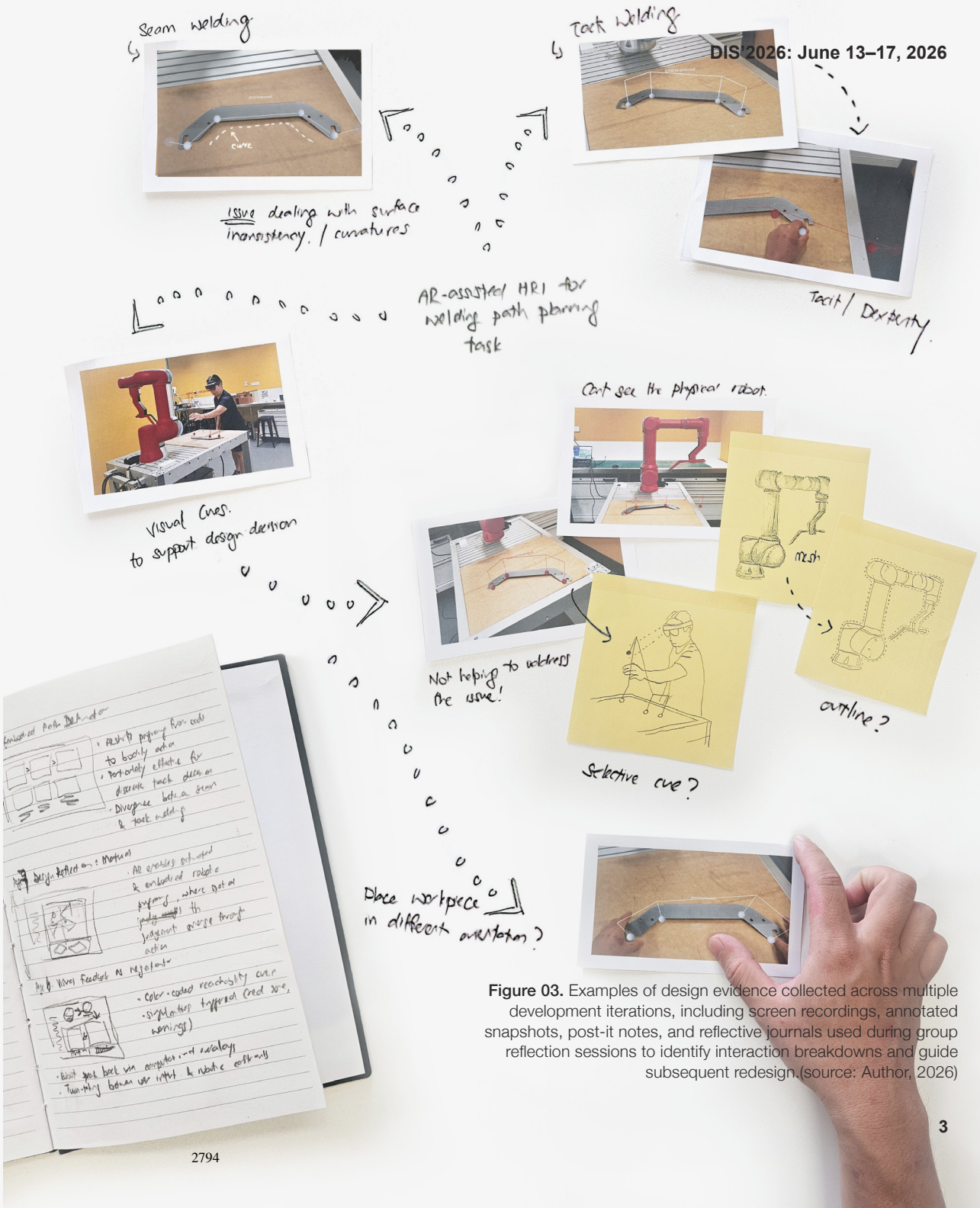


Figure 03. Examples of design evidence collected across multiple development iterations, including screen recordings, annotated snapshots, post-it notes, and reflective journals used during group reflection sessions to identify interaction breakdowns and guide subsequent redesign.(source: Author, 2026)

SYSTEM ARCHITECTURE

The proposed system comprises three tightly coupled layers: a physical fabrication setup, a computational backend, and a spatially anchored AR interface. The physical setup includes a 6 degree-of-freedom (DOF) collaborative robotic arm equipped with a 3D printed simulated welding tool, alongside a metal workpiece positioned within a shared workspace. The Microsoft HoloLens 2 provides a see-through view of the environment, enabling digital content to be overlaid directly onto physical workspace.

At the computational level, Grasshopper, a parametric design environment is used to define welding logic and generate target points [15]. Plugins such as Robots [16] and Fologram [17] support robotic path planning and the creation of a cyber-physical environment, allowing user input such as the placement or adjustment of robotic weld points to be streamed back into the computational environment in real-time. The system evaluates the feasibility for each user input using inverse kinematics (IK) solver. Rather than automatically correcting the invalid inputs, the IK exposes these infeasible configurations as visual feedback to assist user decision-making.

INTERACTION WORKFLOW

The workflow consists of four main steps. The operators first position the metal workpiece within the workspace, then launch Grasshopper definition and Fologram to spatially anchor the digital environment over the physical workspace using a visual marker. This establishes a shared spatial reference between the workpiece, robotic arm, and AR overlay, which is critical for accurate spatial reasoning during welding path definition.

Operators then visualise weld paths and adjust waypoints using pinch-and-drag gestures, with changes reflected immediately through a simplified trajectory preview. Operators can also switch between tack and seam welding modes and adjust key operational parameters via a spatially anchored virtual menu, including (i) offset from the workpiece, (ii) welding angle, and (iii) trajectory resolution (number of welding point).

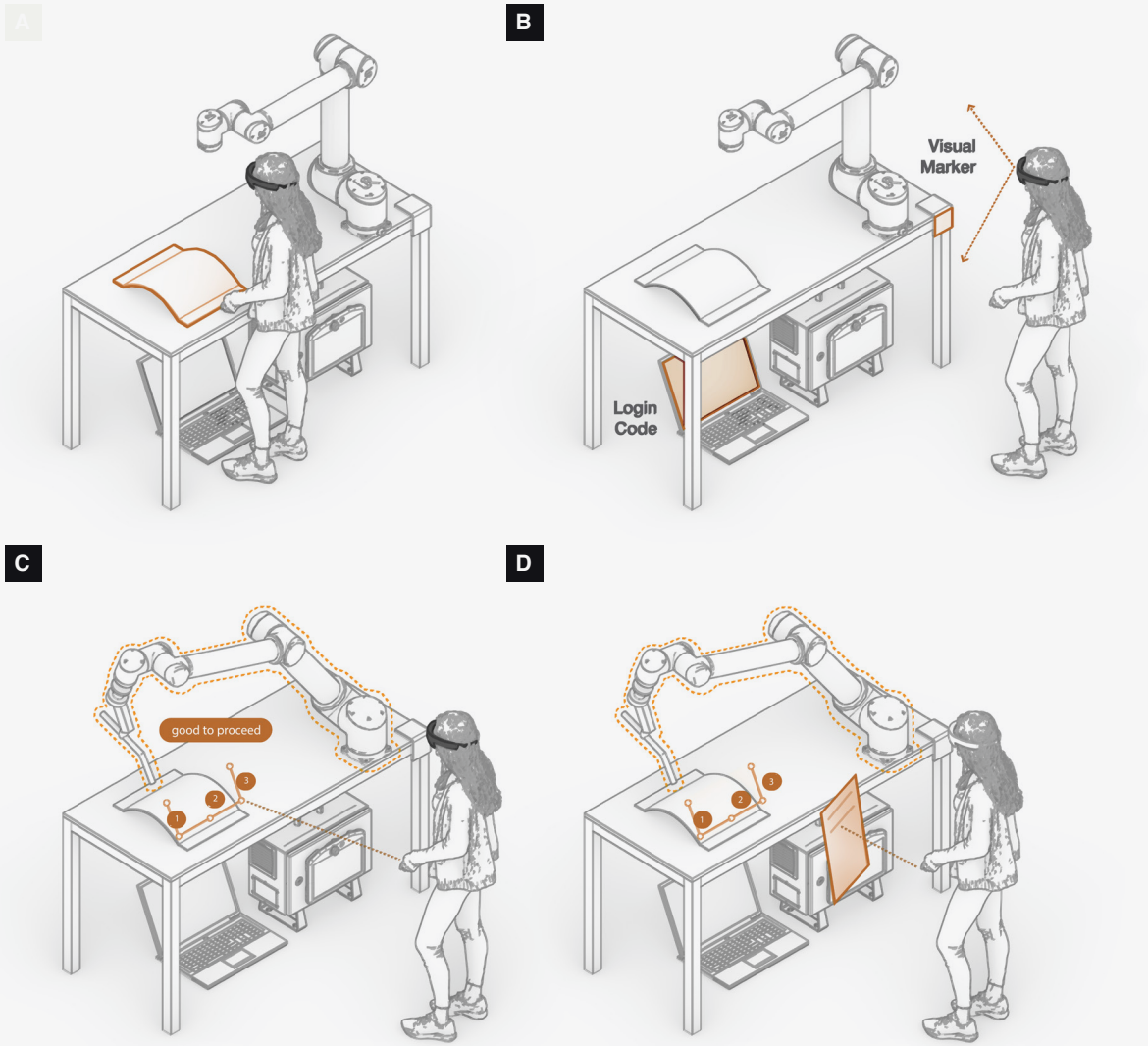


Figure 04. Overall interaction workflow. (A) Placing metal workpiece within the robot operational range; (B) Initialise the computational pipeline; (C) Adjust robotic waypoints to suit; and (D) Modify other design parameters. (source: Author, 2026)

FINDING 1:
SITUATED AND EMBODIED PROGRAMMING

Through this investigation, we observed how AR interaction shifted robotic welding path definition from abstract, code-based control to a situated and embodied activity within the physical workspace. Instead of specifying welding paths through detached interfaces, operators defined seam and tack locations directly on the workpiece using gesture, spatial inspection, and visual feedback as primary interaction modes. This aligns with [6]'s notion of knowledge-through-making, where understanding emerges through material engagement rather than abstract specification.

This embodiment manifested differently across welding tasks. Programming-through-Demonstration (PtD) approaches [16,14,15] appeared well suited to seam welding, where continuous motion captures surface irregularities and curvature (**Figure 05**, bottom right). In contrast, gestural input proved more effective for tack welding, which requires discrete spatial judgement, segmentation, and rapid in-situ decision-making (**Figure 05**, top).

These observations suggest that AR not only supports intuitive input, but also reshapes how operators interpret robotic constraints. Computational feedback became part of a reflective dialogue, allowing users to anticipate, negotiate, and adjust robotic action during path planning. In this sense, the AR interface functioned as a reflective medium through which the robot could "speak back," supporting co-creative decision-making in human–robot collaboration, consistent with prior work on feedback-mediated cobot interaction [13,11].

Design Implication: AR-mediated feedback enables robotic programming to become an embodied activity when interaction is anchored to the physical workspace and directly manipulable in situ.



Figure 05. Mixed reality snapshots of operator(s) using the AR-assisted HRI system for robotic welding path-planning tasks. (A) Tack welding, and (B) Seam welding. (source: Author, 2026)

FINDING 2:
VISUAL FEEDBACK AS NEGOTIATED DIALOGUE

As we experimented with defining welding paths through the AR interface, we notice that visual feedback played a central role in shaping how robotic constraints were understood and addressed. Rather than functioning as error correction or automated validation, feedback appeared as part of the interaction, allowing for interpretation and adjustment of robotic reachability and kinematic limitations during path definition (see **Figure 06**).

As welding points and trajectories were adjusted, the system continuously evaluated feasibility and exposed potential issues through colour-coded indicators, trajectory annotations, and spatial warnings. These cues did not prevent action or resolve conflicts automatically. Instead, they would allow operator(s) to pause, reassess, and adjust waypoints in response to what became visible in the AR overlay. In this way, the robot "speaks back," transforming technical validation into a conversational mechanism that supports reflective decision-making.

However, this early design iteration also revealed critical tension. When multiple forms of visual feedback were presented simultaneously, dense mesh-based overlays and colour-filled regions frequently occluded the physical metal surface. While these representations effectively communicated urgency, they also interfered with precise spatial judgement. In these moments, visual feedback ceased to function as a conversational response and instead became a competing visual layer, breaking the turn-taking rhythm between human and robotic action.

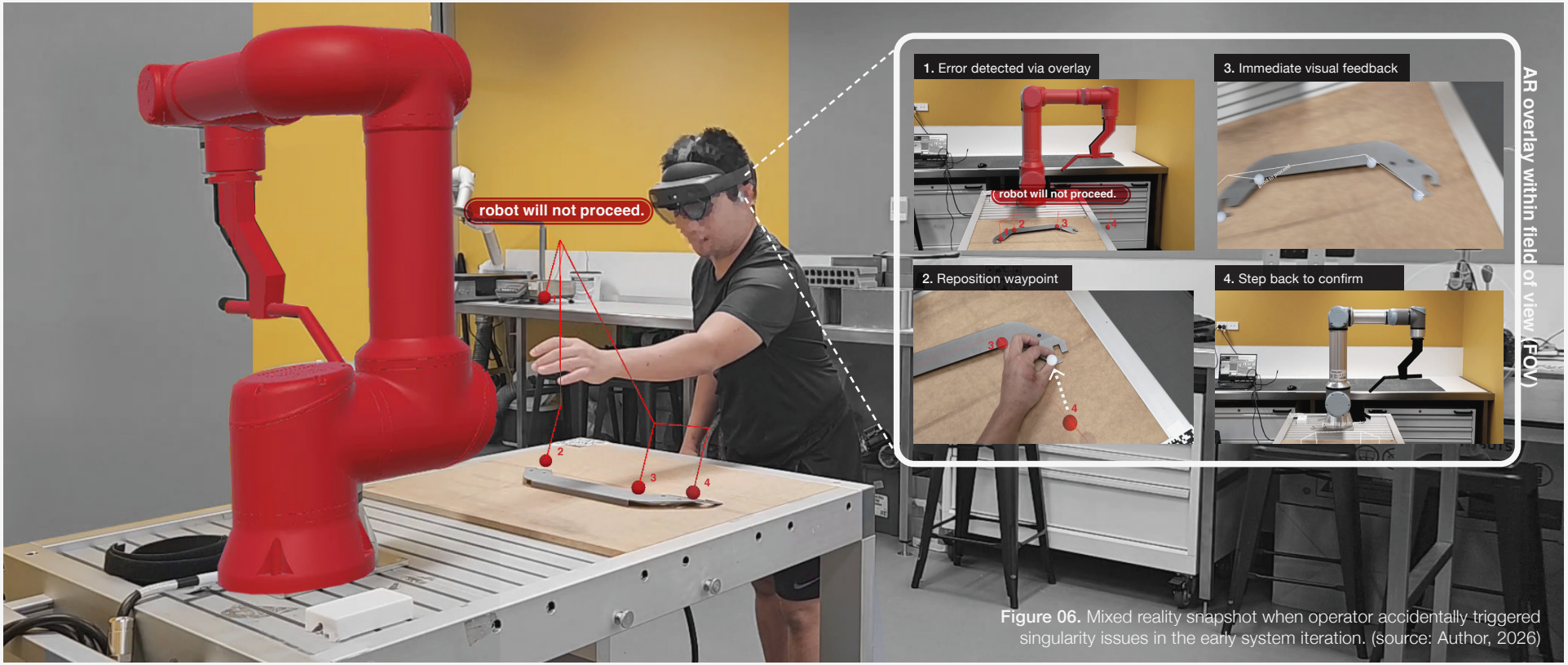


Figure 06. Mixed reality snapshot when operator accidentally triggered singularity issues in the early system iteration. (source: Author, 2026)

FINDING 2: (CONTS.)

VISUAL FEEDBACK AS NEGOTIATED DIALOGUE

In response, the later system iteration introduced a more selective approach to visual feedback. Rather than displaying all reachability and kinematic information continuously, the interface highlighted only the specific waypoints or path segments that triggered robotic constraints. These focused visual cues **A** were accompanied by concise textual annotations **C** that explained the nature of the constraint without introducing additional visual clutter. This approach directed user attention to the actionable source of the issue while preserving situational awareness.

Secondly, we also moved away from the dense mesh-based visualisations in favour of a simplified representation, with outline-based indicators and reduced opacity to minimise occlusion **B**. Furthermore, the robot's outline was hidden by default and only becoming visible when a reachability or kinematic issue arose. This design refinement allowed the system to maintain the communicative role of feedback while minimising occlusion to the physical workspace.

Through reflective iteration, this breakdown was reframed as a design insight rather than a technical limitation. The issue was not incorrect or insufficient feedback, but a lack of selectivity. When multiple visualisations were presented simultaneously, visual feedback ceased to function as a conversational response and instead became a competing visual layer, breaking the turn-taking rhythm between human and robotic action. By comparing these two system iterations, our pictorial demonstrates that how feedback selectivity altered spatial judgement during robotic path planning.

Design Implication: AR visual feedback should remain selective and context-dependent to preserve interpretability and maintain a conversational interaction between human and robot.

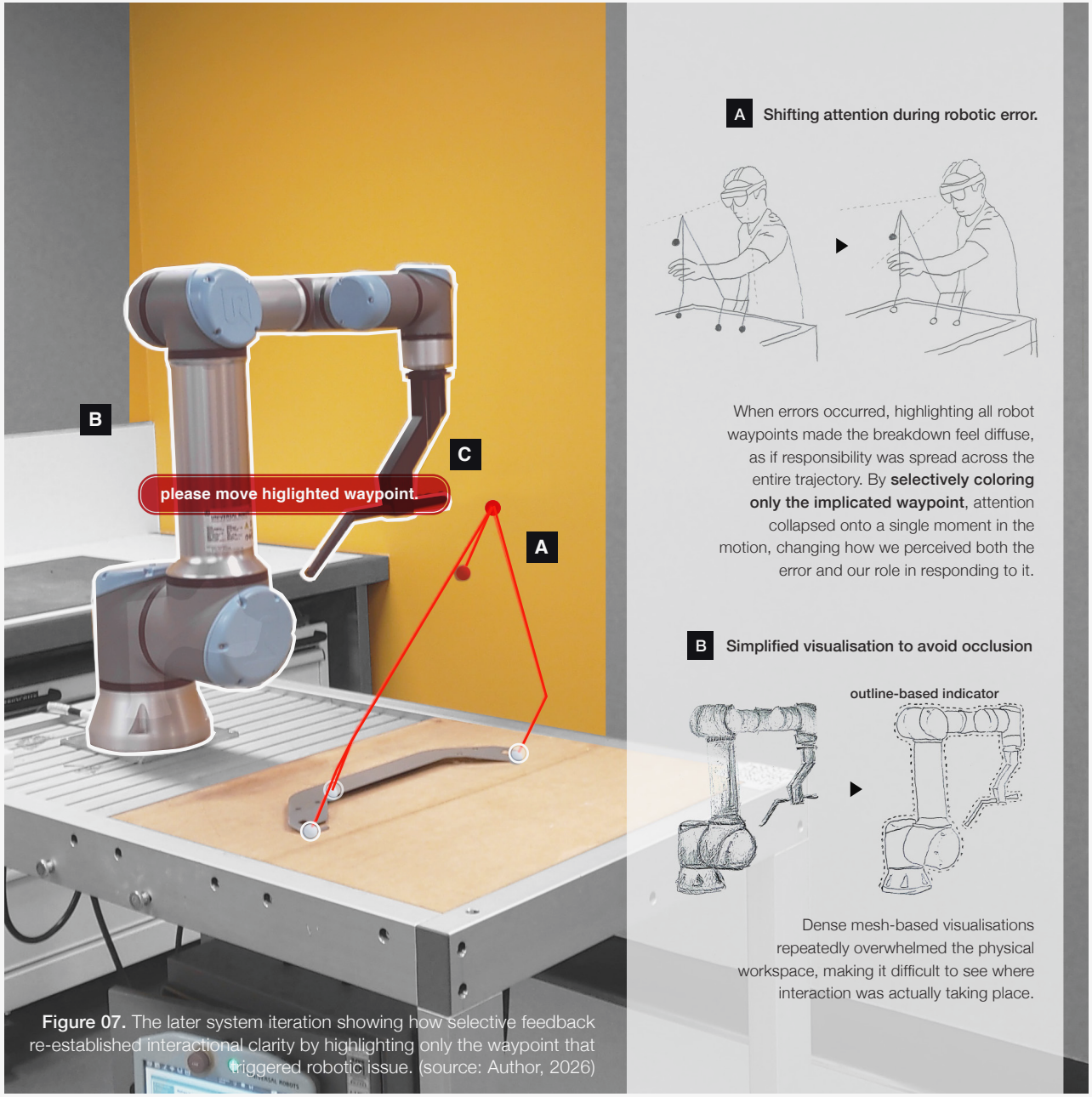


Figure 07. The later system iteration showing how selective feedback re-established interactional clarity by highlighting only the waypoint that triggered robotic issue. (source: Author, 2026)

FINDING 3:
WORKPIECE PLACEMENT AS DESIGN VARIABLE

Another insight emerging from the investigation was that visualising robotic reachability and constraint feedback reframed the placement of the metal workpiece from a fixed preparatory step into an adjustable design parameter. Rather than adapting welding paths to fit an assumed workpiece position, we began to reposition and reorient the workpiece to better align with robotic feasibility (refer to **Figure 08**).

We saw that material placement became entangled with path planning, as small changes in orientation or location significantly altered reachability, end-effector orientation, and the need for corrective feedback. This observation resonates with [2] who emphasise that the orientation of the physical workpiece and its placement within the robotic workspace must be carefully considered to support reachability and manipulability. Our investigation showed that AR interface made these relationships immediately legible, enabling operators to reason about spatial trade-offs directly within the workspace.

This shift highlights how AR-mediated feedback can extend design agency beyond trajectory definition to include material configuration itself. By externalising the robotic constraints in situ, the system supports a more holistic form of spatial reasoning, where material, robot, and action are negotiated together rather than sequentially resolved.

Design Implication: AR feedback extend design agency beyond trajectory definition to include material placement, allowing robot, workspace, and task to be negotiated in real time.



Figure 08. Video screenshot of operator(s) repositioning and reorienting physical workpiece within the robot workspace to ensure it remained within robotic reach. (source: Author, 2026)

FINDING 4:
AR DRIFT AND SPATIAL FIDELITY

During the initial system development, the AR drift surfaced as a recurring issue. This issue is particularly evident when the visual marker is outside the AR device's field of view (FOV) for extended periods. This breakdown revealed a tension between the technical constraints of marker-based tracking and the experiential continuity operators would expect. From our perspective, this momentary loss of alignment not only disrupted the perceived spatial flow of the welding path but also diminished confidence in the AR system.

Rather than treating the AR drift as a calibration issue, our pictorial frames this tracking instability as an "infrastructural breakdown" or interactional breakdown that directly shapes spatial reasoning, and engagement with the robotic constraints. Given that robotic welding demands high spatial precision, even minor registration errors can significantly compromise task accuracy as well as how operators interpret robotic intent and feasibility. This incidence highlights the fragility of shared spatial understanding via AR environment. Furthermore, the need to re-anchor the digital environment mid-task interrupts the conversational flow between the operator and the robotic arm, highlighting how infrastructural conditions can influence interaction quality.

These observations prompted a critical reflection on the system's tracking approach, not merely as a technical infrastructure, but as interaction materials that shape how robotic systems are perceived and trusted. Although the use of a single visual marker supported the initial establishment of a functional cyber-physical workspace, it proved insufficient for tasks that require high precision. Future iterations may explore multi-marker configurations (see **Figure 09**), hybrid marker-SLAM approaches, or integrating periodic recalibration protocols to stabilise the spatial fidelity, aligning with strategies reported by [7,18].

Design Implication: Tracking stability should be treated as part of the interaction design, as spatial drift directly affects trust, interpretation of feedback, and collaboration quality.

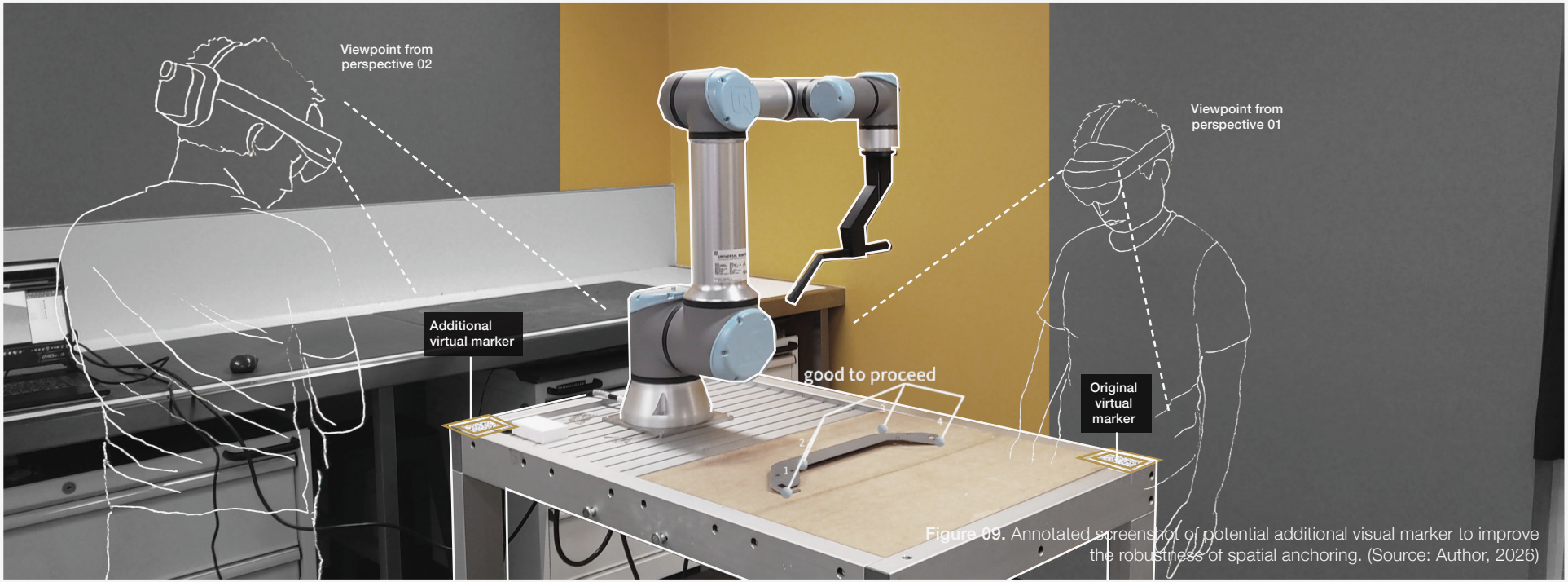


Figure 09. Annotated screenshot of potential additional visual marker to improve the robustness of spatial anchoring. (Source: Author, 2026)

LIMITATIONS AND FUTURE WORK

The study also highlights several limitations and directions for future work. The reflections presented here are based on designer-led iterations in a controlled research setting and do not represent the full range of industrial welding practice. This designer-led exploration was not intended to sidestep user involvement or evaluation, but to examine in finer detail how interaction unfolds and what design implications emerge from it. Future work will further develop, iterate, and evaluate this proposal with professional welders, while also considering how the system performs across different levels of expertise. The interaction qualities explored here may also be relevant to other tacit, skill-based fabrication tasks, such as carving, sculpting, and related material-shaping processes. The current investigation was limited to small workpieces within a fixed robotic workspace. Extending the approach to larger-scale or mobile robotic platforms introduces additional challenges, especially where shared spatial reference frames shift over time. Future research could explore multi-marker tracking or sensor-fusion strategies to support more reliable alignment between human, robot, and workspace.

CONCLUSION

This pictorial presented an AR-assisted robotic welding system developed through a Research-through Design (RtD) approach. The pictorial also provides an account of the embodied, visual, and experiential reasoning underlying our design decisions.

Rather than evaluating efficiency or task performance, our study focused on how AR-mediated interaction reshapes the way humans perceive, negotiate, and respond to robotic constraints during path planning. Through iterative Research through Design (RtD), we found that seam welding aligns well with Programming-through-Demonstration (PtD), given its ability to capture potential surface irregularities, whereas our proposed approach appears more suitable for tack welding tasks, which often require segmentation, reasoning, and iterative adjustment. These findings highlight that different robotic tasks require different interaction strategies. Across iterations, our investigation also reveals that computational feedback does not simply inform correctness but operates as a reflective and conversational mechanism through which the robotic system "spoke back" through selective visual cues that shape situated decision-making and embodied understanding. Instead of introducing a new interaction technique, this pictorial reframes AR-assisted robotic path planning as a reflective, situated activity shaped by feedback and infrastructural breakdowns.

The key contributions of this research are twofold: (1) an RtD-based exploration of an AR-assisted robotic path-planning workflow that supports both tack and seam welding through gestural input and real-time inverse kinematic feedback; and (2) transferable design knowledge for HCI research, showing how feedback selectivity, embodiment, material placement, and infrastructural breakdowns shape spatial reasoning and decision-making in AR-mediated robotic fabrication.

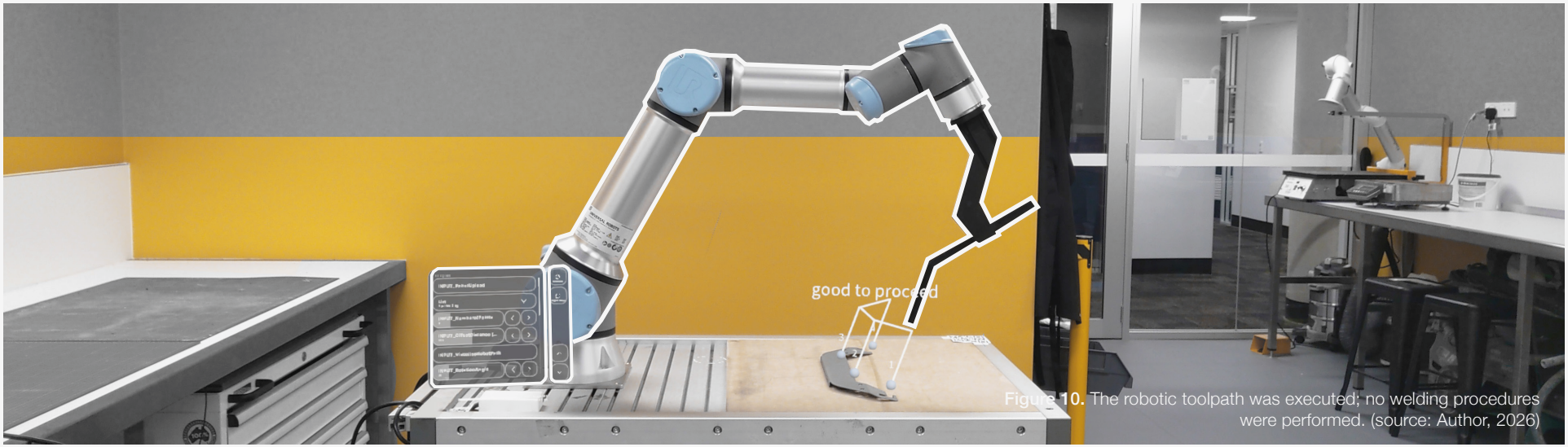


Figure 10. The robotic toolpath was executed; no welding procedures were performed. (source: Author, 2026)

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SUPPLEMENTAL DOCUMENTS

The supplementary documentation is openly available at Github repositories[1]. The files include full-resolution of the pictorial, illustrations, annotated photographs, video clips, codes.

1 https://github.com/LoyWeiWin/Pub_DIS_SupplementaryDocument_DesignofanARHMDInterfaceforInteractiveControlofRoboticWelding

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